Introductory Econometrics

Dummy Variables

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Dummy Variables

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A Single Dummy Variable

- **Qualitative Information**
  - Examples: gender, race, industry, region, rating grade, …
  - A way to incorporate qualitative information is to use dummy variables
  - They may appear as the dependent or as independent variables
- **A single dummy independent variable**

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u$$

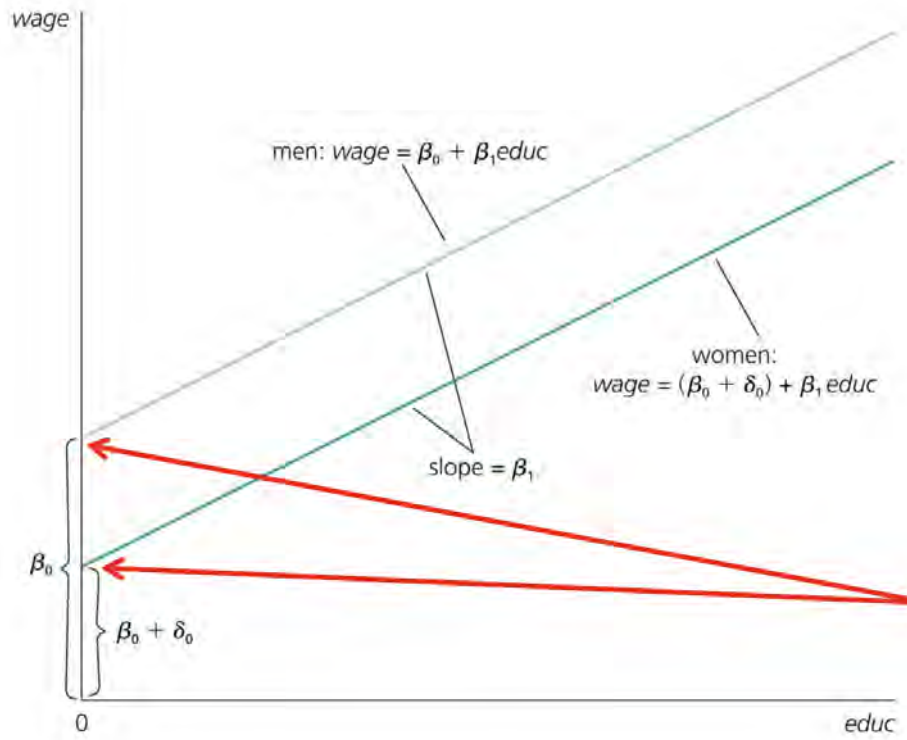


= the wage gain/loss if the person is a woman rather than a man (holding other things fixed)

Dummy variable:
=1 if the person is a woman
=0 if the person is man

A Single Dummy Variable

- **Graphical Illustration**



Alternative interpretation of coefficient:

$$\delta_0 = E(wage|female = 1, educ)$$
$$-E(wage|female = 0, educ)$$

i.e. the difference in **mean** wage between men and women with the same level of education.

Intercept shift

A Single Dummy Variable

To examine the effects of incumbency (running for re-election to a house seat you currently hold) in the 2018 US House of Representatives election, we run the following regression on our dataset of election winners:

$$log(vshare_i) = \beta_0 + \beta_1 Inc_i + \beta_2 D_i + \beta_3 log(share16_i) + U_i, \tag{1}$$

where the dependent variable, $log(vshare)$, is the log of the vote-share the winner of a given seat received in the 2018 election; Inc is a dummy variable with $Inc = 1$ if the winner was an incumbent and $Inc = 0$ if the winner was not the incumbent; D is a dummy variable indicating if the winner was a democrat or republican with $D = 1$ for Democrat and $D = 0$ for Republican (assume everyone in the sample is either a Democrat or Republican); $log(share16)$ is the share of the vote the winner's party received in the district in 2016. Assume $\mathbb{E}[U_i|Inc_i, D_i, log(share16_i)] = 0$. The sample size is $n = 435$. It is large enough to assume all large sample approximations work well.

A Single Dummy Variable

- **Further example: Effects of training grants on hours of training**

Hours training per employee

Dummy indicating whether firm received training grant

$$\widehat{hrsemp} = \underset{(43.41)}{46.67} + \underset{(5.59)}{26.25}\,grant - \underset{(3.54)}{0.98}\,\log(sales)$$

$$- \underset{(3.88)}{6.07}\,\log(employ),\ n = 105, R^2 = .237$$

- **This is an example of <u>program evaluation</u>**
  - Treatment group (= grant receivers) vs. control group (= no grant)
  - Is the effect of treatment on the outcome of interest <u>causal</u>? We hope we have controlled for as many factors as possible that might be related to whether a firm received a grant and to its levels of training.

A Single Dummy Variable

- **Using dummy explanatory variables in equations for log(y)**

  Example: Housing price regression (HPRICE1.dta)

$$\widehat{\log}(price) = \underset{(.65)}{-1.35} + \underset{(.038)}{.168}\ \log(lotsize) + \underset{(.093)}{.707}\ \log(sqrft)$$

$$+ \underset{(.029)}{.027}\ bdrms + \underset{(.045)}{.054}\ colonial$$

Dummy indicating whether house is of colonial style

$$n = 88, R^2 = .649$$

$$\Rightarrow \quad \frac{\Delta \log(price)}{\Delta colonial} \approx \frac{(1/100)\%\Delta price}{\Delta colonial} = 0.054$$

As the dummy for colonial style changes from 0 to 1, the house price increases by 5.4%

A Single Dummy Variable

- **Example: Wage equation**

$$\widehat{\log(wage)} = \underset{(.099)}{.417} - \underset{(.036)}{.297}\, female + \underset{(.007)}{.080}\, educ + \underset{(.005)}{.029}\, exper$$

$$- \underset{(.00010)}{.00058}\, exper^2 + \underset{(.007)}{.032}\, tenure - \underset{(.00023)}{.00059}\, tenure^2$$

$$n = 526, R^2 = .441$$



As the dummy for female changes from 0 to 1, the hourly wage decreases by 29.7%

Dummy Variables for Multiple Categories

- **Using dummy variables for multiple categories**

  1) Define membership in each category by a dummy variable

  2) Leave out one category (which becomes the base category)

- **Example: Wage equation with gender and marriage status**
  - Two qualitative variables: *married* and *female*
  - Four categories:

    marrmale (married males): *married* =1, *female* =0

    marrfem (married females): *married* =1, *female* =1

    singmale (single males): *married* =0, *female* =0

    singfem (single females): *married* =0, *female* =1

Dummy Variables for Multiple Categories

- **Example: Wage equation (cont.)**

$$\widehat{\log}(wage) = \underset{}{.321} + \underset{(.100)}{.213}\, marrmale - \underset{(.058)}{.198}\, marrfem - \underset{(.056)}{.110}\, singfem + \underset{(.007)}{.079}\, educ + \underset{(.005)}{.027}\, exper - \underset{(.00011)}{.00054}\, exper^2 + \underset{(.007)}{.029}\, tenure - \underset{(.00023)}{.00053}\, tenure^2$$

Holding other things fixed, married women earn 19.8% less than single men (= the base category)

- What if we want to compare single and married women? Whether the estimated difference between single and married women is statistically significant?

Dummy Variables for Multiple Categories

- **Incorporating ordinal information using dummy variables**
- **Example: City credit ratings and municipal bond interest rates**

Municipal bond rate

Credit rating from 0-4 (0=worst, 4=best)

$$MBR = \beta_0 + \beta_1 CR + other\ factors$$

This specification would probably not be appropriate as the credit rating only contains ordinal information. A better way to incorporate this information is to define dummies:

$$MBR = \beta_0 + \delta_1 CR_1 + \delta_2 CR_2 + \delta_3 CR_3 + \delta_4 CR_4 + other\ factors$$

Dummies indicating whether the particular rating applies, e.g. CR_1=1 if CR=1 and CR_1=0 otherwise. All effects are measured in comparison to the worst rating (= base category).

Dummy Variables for Multiple Categories

- What if the ordinal variable takes on too many values?
  - we can break it down into categories
- **Example: effects of law school rankings on starting salaries** (LAWSCH85.dta)

$$\widehat{\log}(salary) = \underset{(.41)}{9.17} + \underset{(.053)}{.700}\, top10 + \underset{(.039)}{.594}\, r11to25 + \underset{(.034)}{.375}\, r26to40$$
$$+ \underset{(.028)}{.263}\, r41to60 + \underset{(.021)}{.132}\, r61to100 + \underset{(.0031)}{.0057}\, LSAT$$
$$+ \underset{(.074)}{.014}\, GPA + \underset{(.026)}{.036}\, log(libvol) + \underset{(.0251)}{.0008}\, log(cost)$$
$$n = 136, R^2 = .911, \bar{R}^2 = .905$$



Holding other things fixed, students from top 10 schools earn 70% more than from schools below 100

- Whether breaking the rank into different groups is an improvement?

Interactions Involving Dummy Variables

- **Interactions Among Dummy Variables**
- **Example: Wage equation with gender and marriage status**

  We can achieve the same thing by using an interaction between *female* and *married*.

$$\widehat{lwage} = \underset{(.100)}{.321} - \underset{(.056)}{.110}\, female + \underset{(.055)}{.213}\, married - \underset{(.072)}{.301}\, female \cdot married + \ldots$$

$$n = 526,\ R^2 = .461$$

- For single women, *female* = 1 and *married* = 0, the intercept is

  $.321 - .110 = .211$

- The intercept for married women is the sum of all coefficients

  $.321 - .110 + .213 - .301 = .123$

Interactions Involving Dummy Variables

Part III

Now consider a nonlinear specification of the model

$$log(vshare_i) = \beta_0 + \beta_1 Inc_i + \beta_2 D_i + \beta_3 log(share16_i) + \beta_4 (Inc_i \times D_i) + U_i. \quad (2)$$

12. (5') What is the main advantage of this new regression compared with the one in part I? Explain how this model improves upon the model in equation (1).

13. (6') Write out the incumbency advantage for a Republican and for a Democrat as functions of $\{\beta_0, \beta_1, \beta_2, \beta_3, \beta_4\}$ under the model given by (2).

Interactions Involving Dummy Variables

- **Interactions involving dummy variables**
- **Allowing for different slopes**

Interaction term

$$\log(wage) = \beta_0 + \delta_0 female + \beta_1 educ + \delta_1 female \cdot educ + u$$

β_0 = intercept men

β_1 = slope men

$\beta_0 + \delta_0$ = intercept women

$\beta_1 + \delta_1$ = slope women

- **Interesting hypotheses**

$H_0 : \delta_1 = 0$

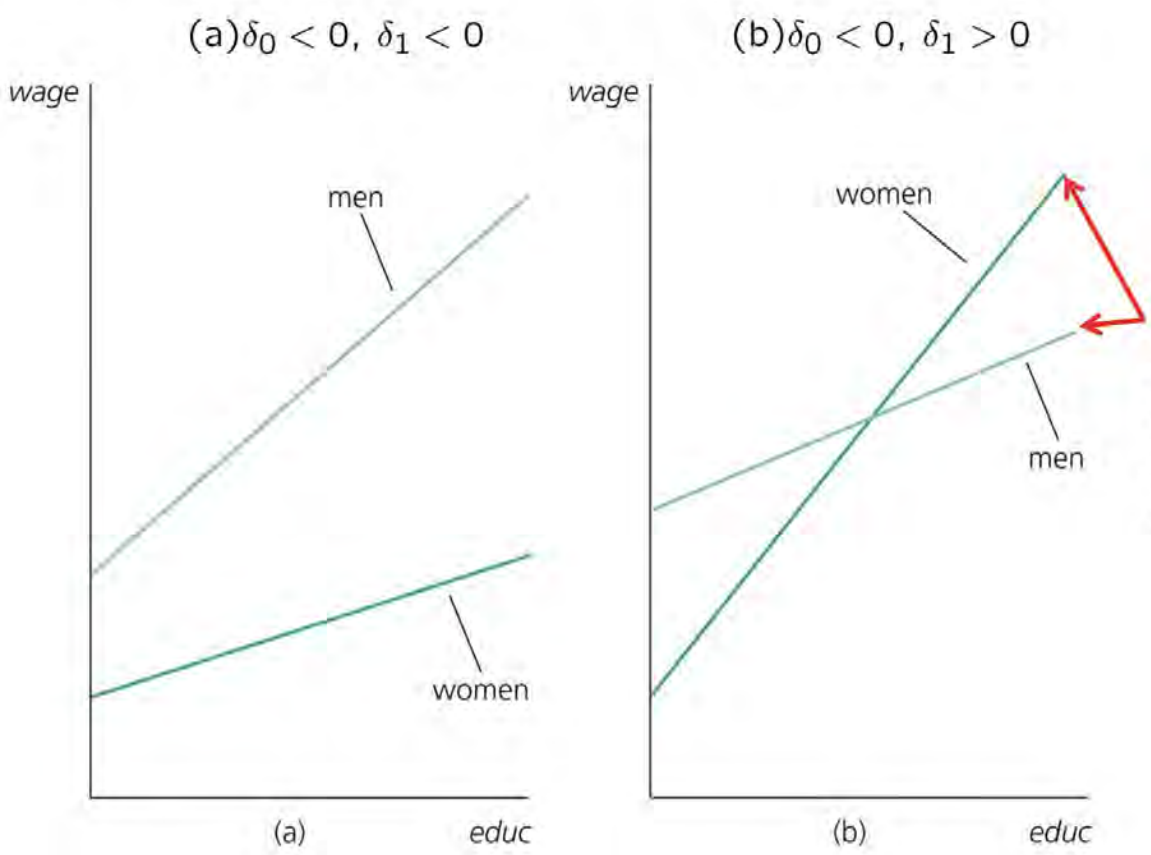
The return to education is the same for men and women

$H_0 : \delta_0 = 0, \delta_1 = 0$

The whole wage equation is the same for men and women

Interactions Involving Dummy Variables

- **Graphical illustration**



Interacting both the intercept and the slope with the female dummy enables one to model completely independent wage equations for men and women

Interactions Involving Dummy Variables

- **Estimated wage equation with interaction term**

$$\widehat{\log(wage)} = \underset{(.119)}{.389} - \underset{(.168)}{.227}\, female - \underset{(.008)}{.082}\, educ$$

$$- \underset{(.0131)}{.0056}\, female{\cdot}educ + \underset{(.005)}{.029}\, exper - \underset{(.00011)}{.00058}\, exper^2$$

$$+ \underset{(.007)}{.032}\, tenure - \underset{(.00024)}{.00059}\, tenure^2, n = 526, R^2 = .441$$

No evidence against hypothesis that the return to education is the same for men and women

Does this mean that there is no significant evidence of lower pay for women at the same levels of educ, exper, and tenure? No: we have to test the joint significance on both female dummy and the interaction term.

Interactions Involving Dummy Variables

- **Testing for differences in regression functions across groups**
- **Example: whether the same regression model describes college GPA for male and female college athletes.** (GPA3.dta)
- **Restricted model (same regression for both groups)**

College grade point average

Standardized aptitude test score

High school rank percentile

$$cumgpa = \beta_0 + \beta_1 sat + \beta_2 hsperc + \beta_3 tothrs + u$$

Total hours spent in college courses

- **Unrestricted model (contains full set of interactions)**

$$cumgpa = \beta_0 + \delta_0 female + \beta_1 sat + \delta_1 female \cdot sat + \beta_2 hsperc$$

$$+ \delta_2 female \cdot hsperc + \beta_3 tothrs + \delta_3 female \cdot tothrs + u$$

Interactions Involving Dummy Variables

- **Null hypothesis**

$$H_0 : \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0$$

All interaction effects are zero, i.e. the same regression coefficients apply to men and women

- **Estimation of the unrestricted model**

$$\widehat{cumgpa} = \underset{(.21)}{1.48} - \underset{(.411)}{.353}\, female + \underset{(.0002)}{.0011}\, sat + \underset{(.00039)}{.00075}\, female \cdot sat$$

$$- \underset{(.0014)}{.0085}\, hisperc - \underset{(.00316)}{.00055}\, female \cdot hisperc$$

$$+ \underset{(.0009)}{.0023}\, tothrs - \underset{(.00163)}{.00012}\, female \cdot tothours$$

Tested individually, the hypothesis that the interaction effects are zero cannot be rejected. But here we should look at the F-statistic

Interactions Involving Dummy Variables

- **Joint test with F-statistic**

Null hypothesis is rejected

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{(85.515 - 78.355)/4}{78.355/(366-7-1)} \approx 8.18$$

- **Alternative way to compute F-statistic: <u>Chow-Test</u>**

1) Pool the data and estimate a single regression. This is the restricted model, and produces the restricted SSR. Call this the pooled SSR, SSR_p

2) Estimate the regressions on the two groups (say, 1 and 2) separately. Get the SSR_1 and SSR_2. The unrestricted SSR is $SSR_1 + SSR_2$ (and this is the same as the regression that includes the full set of interactions).

Interactions Involving Dummy Variables



- **Chow-Test**
- **The *F* statistic is:**

$$F = \frac{[SSR_p - (SSR_1 + SSR_2)]/(k+1)}{(SSR_1 + SSR_2)/[n - 2(k+1)]}$$

Chow-Test only use the restricted model

SSR_p: regression on pooled data (groups 1 and 2)

SSR_1: regression on group 1

SSR_2: regression on group 2

k: number of independent variables in the restricted model

n: number of observations